Enthusiast Course COURSE

PHASE : Speed & Accuracy Test

TARGET : PRE-MEDICAL 2023

Test Type :**PRACTICE TEST** Test Pattern :**NEET (UG)**

TEST DATE : ________

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	3	1	3	3	3	1	4	2	3	4	3	4	3	1	3	3	3	2	2	2	2	1	3	4	3	2	1	3	4	1
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45															
A.	2	1	3	3	2	2	3	3	2	1	2	1	3	2	2															

HINT – SHEET

1. Ans (3)

$v_{in} = 10v$; KVL in i/p loop

$v_{BE} = 0v \quad v_{in} = i_B R_b + v_{BE}$

$v_{CE} = 0 \quad v_{in} = i_B R_b + v_{BE}$

$i_B = \frac{10}{400} \times 10^{-3} = 0.025mA$

KVL in o/p loop

$V_{CC} = i_C R_C + V_{CE}$

$\Rightarrow 10 = i_c\, 3 \times 10^3$

$i_c = 10/3\times10^3 = 3.33mA$

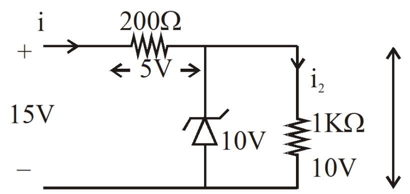
2. Ans (1)

for –ve half cycle diode is RB hence o/p across resistance is zero (OV)

+ve half cycle diode is FB hence o/p across resistance is +1v

3. Ans (3)

$v_{in} > v_z$ hence diode is at B/D



$i = 5/200 = 0.025A = 25mA$

$i_L = 10/10^3 = 10mA$

$i_Z = i - i_L = 25 - 10 = 15mA$

5. Ans (3)

$n_i^2 = n_e \times n_h$

$\Rightarrow n_e = \frac{n_i^2}{n_h} = \frac{\left(0.6 \times 10^{12}\right)^2}{4 \times 10^{20}}$

$= \frac{0.36 \times 10^{24}}{4 \times 10^{20}} = 0.09 \times 10^4$

$= 9 \times 10^2\ m^{-3}$

6. Ans (1)

current gain (β) =

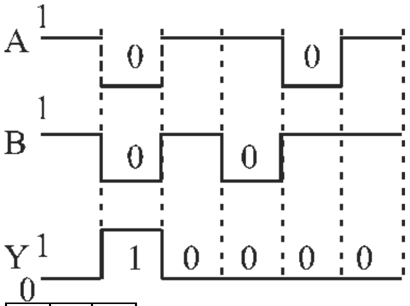
$\frac{\Delta I_C}{\Delta I_B} = \frac{(15-10) \times 10^{-3}}{(150-50) \times 10^{-6}}$

$= \frac{5 \times 10^{-3}}{100 \times 10^{-6}} = \frac{5000}{100} = 50$

7. Ans (4)

In reverse bias –ve terminal to P side & +ve terminal to N side , with increase in R.B potential width of depletion layer increase.

10. **Ans (4)**



A	B	Y
1	1	0
0	0	1
1	1	0
1	0	0
0	1	0

11. **Ans (3)**

$A_V = \beta \dfrac{R_L}{R_i} \left\{\because \beta = A_V \dfrac{R_i}{R_L}\right\}$

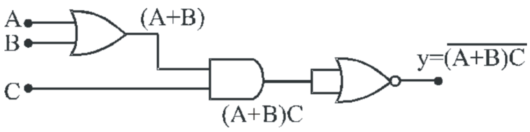
$A_P = \beta A_v$

$A_P = A_V^2 \dfrac{R_i}{R_L} = (25)^2 \times \dfrac{50}{250} = \dfrac{625}{5} = 125$

12. **Ans (4)**

Transistor is used as an inverter or switch b/w (I) & (III) region of V.I curve.

13. **Ans (3)**



put A = 0 B = 1 C = 0

y = 1

put A = 0 ; B = 0 ; C = 1

y = 1

14. **Ans (1)**

$\dfrac{hc}{\lambda} \geqslant Eg$

$\Rightarrow \lambda \dfrac{hc}{Eg} = \dfrac{12400}{1.8} = 6,888\,\mathring{A}$

15. **Ans (3)**

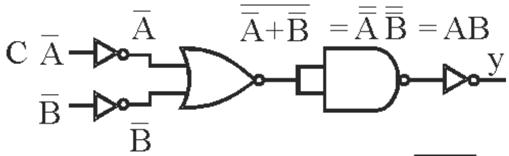
$\Delta I_B = (200 - 150) \times 10^{-6} = 50 \times 10^{-6}$ A

$\Delta I_C = (15 - 10) \times 10^{-3} = 5 \times 10^{-3}$ A

$\beta = \dfrac{\Delta I_C}{\Delta I_B} = \dfrac{5 \times 10^{-3}}{50 \times 10^{-6}} = \dfrac{5 \times 10^{3}}{50} = \dfrac{5000}{50} = 100$

$A_V = \beta \dfrac{R_C}{R_B} = 100 \times \dfrac{20 \times 10^3}{1 \times 10^3} = 2000$

16. **Ans (3)**



$y=\overline{\overline{A.B}}$

$y=A.B$

AND Gate

17. **Ans (3)**

hv = 2.6eV ;

$v = \dfrac{2.6 \times 1.6 \times 10^{-19}}{6.6 \times 10^{-34}}$

$= 0.630 \times 10^{15}$

$= 6.3 \times 10^{14}$ Hz

18. **Ans (2)**

$V_0 = A\left(V_i + \dfrac{10}{100}V_0\right)$ $\{\because V_0 = 50\, V_i\}$

$50V_i = A\left(V_i + \dfrac{50}{10}V_0\right)$

$50V_i = A6V_i$

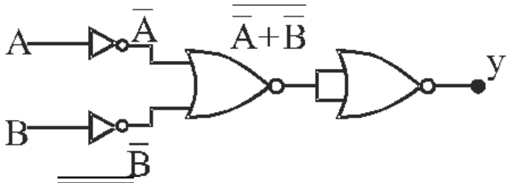
$A = \dfrac{50}{6} = 8.33$

19. **Ans (2)**

Power gain $A_P = \beta Av = A_V^2 \dfrac{R_i}{R_L}$

$P_V = (100)^2 \times \dfrac{50}{200} = \dfrac{100 \times 100}{4} = 2500$

20. **Ans (2)**



$y = \overline{\overline{\bar{A}+\bar{B}}} = \bar{A}+\bar{B} = \overline{A \cdot B}$

NAND gate

21.

Ans (2)

$$\beta = \frac{\Delta I_C}{\Delta I_B}$$

$$= \frac{(12-8)\times 10^{-3}}{(300-250)\times 10^{-6}}$$

$$= \frac{4\times 10^{-3}}{50\times 10^{-6}}$$

$$= \frac{4}{50}\times 10^{3}$$

$$= \frac{4000}{50} = 80$$

22. **Ans (1)**

The kinetic energy of the most energetic electrons is

$$K = \frac{hc}{\lambda} - \phi$$

$$= \frac{1242 \text{ eV nm}}{450 \text{ nm}} - 2.0 \text{ eV}$$

$= 0.76$ eV $= 1.2 \times 10^{-19}$ J

The linear momentum = $m\upsilon$

$$= \sqrt{2mK}$$

$$= \sqrt{2\times(9.1\times 10^{-31}\text{kg})\times(1.2\times 10^{-19}\text{J})}$$

$= 4.67 \times 10^{-25}$ kgms^{-1}

When a charged particle is sent perpendicular to a magnetic field, it goes along a circle of radius

$$r = \frac{m\upsilon}{qB}$$

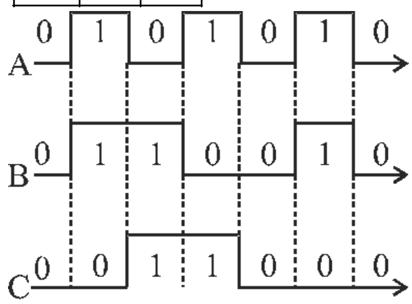
Thus, $0.20 \text{ m} = \dfrac{4.67\times 10^{-25}\text{kg ms}^{-1}}{(1.6\times 10^{-19}\text{C})\times B}$

or, $B = \dfrac{4.67\times 10^{-25}\text{kg ms}^{-1}}{(1.6\times 10^{-19}\text{C})\times(0.20\text{ m})}$

$= 1.46 \times 10^{-5}$ T

23. **Ans (3)**

A	B	C
0	0	0
0	1	1
1	0	1
1	1	0



X-OR Gate

24. **Ans (4)**

Semiconductors have negative temp. coefficient

temp ↑ ρ ↓ σ ↑

temp ↓ ρ ↑ σ ↓

25. **Ans (3)**

Solar cell is always unbiased

27. **Ans (1)**

$V_{dc} = \dfrac{V_0}{\pi} = \dfrac{25}{\pi}$ (for H.W.R.)

28. **Ans (3)**

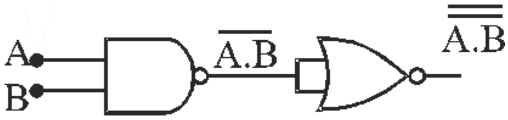
NOR gate

X	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

29. **Ans (4)**

By increase RB pot^n , effective barrier potential increases, thus width of depletion layer also increase.

30. **Ans (1)**



$\Rightarrow$ y= A.B AND gate

31. **Ans (2)**

In HWR ripple frequency is equal to source frequency.

32. **Ans (1)**

(depletion layer) depends only of type of biasing

36. **Ans (2)**

HWR $V_{rms} = \frac{V_m}{2} = \frac{200}{2} = 100v$

37. **Ans (3)**

Final P.D. across capacitor

= peak value of the source voltage

$= 200\sqrt{2} = 283\text{ volt}$

38. **Ans (3)**

$h\upsilon = \frac{hc}{\lambda}J = \frac{hc}{e\lambda}eV$

$= \left(\frac{12400}{\lambda}\overset{\circ}{A}\right)eV = \frac{124}{248}$

= 0.5 eV

42. **Ans (1)**

$\beta = \frac{\alpha}{1-\alpha} = \frac{0.96}{1-0.96} = 24$

$\because I_C = \frac{V_C}{R} = \frac{0.5}{800} = 0.625mA$

$I_B = \frac{I_C}{\beta}$

$= \frac{0.625 \times 10^{-3}}{24}$

$= 26 \times 10^{-6} = 26\mu A$

44. **Ans (2)**

$\alpha = 0.98$ and $\frac{R_{out}}{R_{in}} = 500$

$\therefore \quad \beta = \frac{\alpha}{1-\alpha} = \frac{0.98}{1-0.98} = 49$

Voltage gain

$= \beta\frac{R_{out}}{R_{in}} = 49 \times 500 = 24500$

Power gain

$= \beta^2\frac{R_{out}}{R_{in}} = (49)^2 500 = 1200500$